FootyEdge AI — Unified Production Betting Intelligence Architecture (Master Document)

---

1. Executive Summary

FootyEdge AI is a **betting intelligence system**, not a prediction tool.

It converts football data into:

- probabilistic match outcomes
- expected value (EV) betting opportunities
- risk-adjusted staking decisions
- backtested, performance-validated signals

Primary objective:

Sustained positive expected value (EV) and bankroll growth, not prediction accuracy.

---

2. System Philosophy (Non-Negotiable Rules)

Core Principle

- Deterministic systems first
- ML models second
- LLMs last (only for explanation)

Strict Separation of Concerns

❌ Forbidden in prediction pipeline

- LLM reasoning
- prompt-based probability generation
- AI-generated betting decisions
- probabilistic “opinions”

✅ Allowed in prediction pipeline

- Poisson models
- LightGBM / XGBoost
- deterministic feature engineering
- EV mathematics
- Kelly staking

✅ LLM usage ONLY for:

- human-readable explanations
- betting summaries
- strategy chat assistant
- UI copilots

---

3. Core System Architecture

Data Layer

↓

Feature Engineering Engine

↓

Prediction Engine (Poisson + ML)

↓

Hybrid Probability Layer

↓

Market Intelligence Layer (EV + Sharp Detection + CLV)

↓

Risk Engine (Kelly + bankroll)

↓

Signal Engine

↓

API / Telegram / UI Distribution

↓

Backtesting + Training Pipeline (parallel system)

---

4. DATA LAYER

Sources

- API-Football
- Odds APIs
- Supabase historical matches
- Local datasets (CSV 2001–2025)

Normalized Match Schema

{

"fixture_id": 123,

"home_team": "Arsenal",

"away_team": "Chelsea",

"date": "2026-06-14",

"league": "Premier League"

}

Rules

- Never pass raw API data downstream
- Always normalize first
- Cache all responses (TTL: 3600s)

---

5. FEATURE ENGINEERING ENGINE

Core Feature Groups

Team Form

- last 5 / 10 / 20 matches
- win rate
- draw rate
- goals scored avg
- goals conceded avg

Home/Away Strength

- home_win_rate
- away_win_rate
- home_xG
- away_xG

Head-to-Head

- h2h_home_wins
- h2h_draws
- h2h_away_wins

League Strength

- league rank
- goal difference
- points per match

Player Impact

- injury score
- availability score
- key scorer contribution

🆕 Upgrade Layer (IMPORTANT)

Rolling xG Features

- 5-match window
- 10-match window
- 30-match window

Momentum Index

- attacking pressure trend
- recent scoring intensity

Pressure Index

- goals conceded trend
- defensive instability

Game State Encoding

- winning
- drawing
- losing

---

6. REALTIME STATE ENGINE (NEW CORE MODULE)

Purpose

Convert live match events into structured time-series state vectors.

File

`data/realtime_state_engine.py`

Output State Vector

MatchState:

- minute

- home_xg

- away_xg

- momentum_home

- momentum_away

- pressure_home

- pressure_away

- game_state

Function

- ingests live events
- updates match state per minute
- feeds prediction + market engine

---

7. PREDICTION ENGINE

Base Models

1. Poisson Model

- score simulation up to 12 goals
- generates:

- home/draw/away probabilities
- goal distributions

2. ML Model (LightGBM / XGBoost)

- trained on engineered features
- outputs:
  - win/draw/loss probabilities

🆕 Hybrid Model

final_prob =

0.5 * ML

+ 0.3 * Poisson

+ 0.2 * momentum adjustment

Momentum Adjustment

momentum = home_momentum - away_momentum

adjustment = momentum * 0.1

---

8. MARKET INTELLIGENCE ENGINE

Core Formula

EV = (Probability × Odds) − 1

Tiering

- EV > 0.03 → Value
- EV > 0.07 → Strong
- EV > 0.12 → Premium

🆕 Enhancements

Closing Line Value (CLV)

CLV = (current_odds - opening_odds) / opening_odds

Bookmaker Clustering

- sharp books (low odds deviation)
- soft books (high odds deviation)

Probability Calibration

Platt-style smoothing:

p_calibrated = 1 / (1 + e^(-5(p - 0.5)))

---

9. RISK ENGINE (KELLY SYSTEM)

Formula

f = (bp - q) / b

Where:

- b = odds - 1
- p = probability
- q = 1 - p

Constraints

- max stake = 25% bankroll
- fractional Kelly recommended (≤ 50%)

---

10. VALUE BET ENGINE

Workflow

Raw Probability

↓

Calibration

↓

Momentum Adjustment

↓

EV Calculation

↓

Tier Classification

---

11. SHARP MONEY DETECTOR

Logic

if odds_drop > 8%:

    signal = "Sharp Money Inflow"

Outputs:

- signal strength
- movement classification

---

12. BACKTESTING ENGINE

Requirements

- minimum 5,000 historical matches
- same logic as live system

Metrics

- ROI
- Yield
- Win rate
- Max drawdown
- Profit factor

---

13. SIGNAL ENGINE

Output Types

- Free signals (low EV)
- Premium signals (EV > 0.07)
- Elite signals (EV > 0.12 + sharp confirmation)

Example Output

Arsenal vs Chelsea

Market: Over 2.5 Goals

EV: 12%

Odds: 2.10

Recommended Stake: 2.5% bankroll

Tier: Premium 🔥

---

14. TELEGRAM DISTRIBUTION LAYER

Flow

Prediction

→ Value Engine

→ Kelly Staking

→ Signal Formatter

→ Telegram Bot API

15. ML TRAINING PIPELINE

Directory

ml_pipeline/

  dataset_builder.py

  train.py

  evaluate.py

  export.py

Workflow

Data → Features → Train → Evaluate → Export Model

16. DATABASE DESIGN

Tables

- teams
- matches
- predictions
- value_bets
- user_bets
- model_runs

17. MONITORING SYSTEM

Track:

- ROI
- prediction accuracy
- API errors
- telegram delivery
- bankroll growth

---

18. SAAS LAYER

Plans

Free

- 3 signals/day

Premium

- full signals
- Kelly stakes
- sharp alerts

Elite

- API access
- portfolio tracking

---

19. DEPLOYMENT STRATEGY

Phase 1

- single FastAPI server
- Supabase

- Telegram bot

Phase 2

- split services:
    - prediction-service
    - market-service
    - signal-service

Phase 3

- containerization (Docker)

Phase 4

- Kubernetes (only if scale > 100k users)

---

20. CRITICAL SYSTEM RULES

DO NOT

- use LLMs for prediction
- replace Poisson prematurely
- skip backtesting
- deploy without EV validation

MUST HAVE

- deterministic probability system
- ML + Poisson hybrid
- EV-first decision layer
- Kelly staking before signal release

---

21. STRATEGIC POSITIONING

FootyEdge is:

A sports market inefficiency engine, not a prediction app.

Core advantage:

- probability modeling
- EV extraction
- bankroll optimization
- backtested signal generation

---

22. SUCCESS METRIC

NOT:

- prediction accuracy

YES:

- positive EV
- sustained ROI
- controlled drawdown
- bankroll growth

END OF DOCUMENT